

Task 3

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SA simulation

Weight Scalarization Strategy

The simulated_annealing program is based on a mono-objective algorithm. We have a scalarization approach for transforming the problem (bi-objective) into a function mono-objective. To generate the pareto front, 200 runs are made with the weight uniformly distributed between 0.005 and 0.995. A scale of 10000 is applied at the displacement for normalizing both objectives to comparable orders of magnitude. The stress constraints are enforced through a penalty method is generated by a penalty method in the program for the optimization with a final verification of the final solution. This double approach guaranteed that all the Pareto front respect the structural constraints.

Scalar Energy function :

$$E(x) = w_1 \cdot f_1(x) + w_2 \cdot (\text{scale_f2} \times f_2(x)) + \sum_{i=1,2} 10^6 \cdot \max(0, g_i(x))$$

where :

- w_1, w_2 : scalar weight with $w_1 + w_2 = 1$ et $w_1 \in [0.005, 0.995]$
- $f_1(x)$: structural weight (kg)
- $f_2(x)$: displacement (m)
- $\text{scale_f2} = 10000$: the objective normalize factor
- $g_i(x) \leq 0$: Stress constraints (tension g_1 et compression g_2)

Weight distribution

To generate a Paret front we do 200 independent runs with some weight uniformly distributed:

$$w_1 \in [0.005, 0.995] \text{ avec } w_2 = 1 - w_1$$

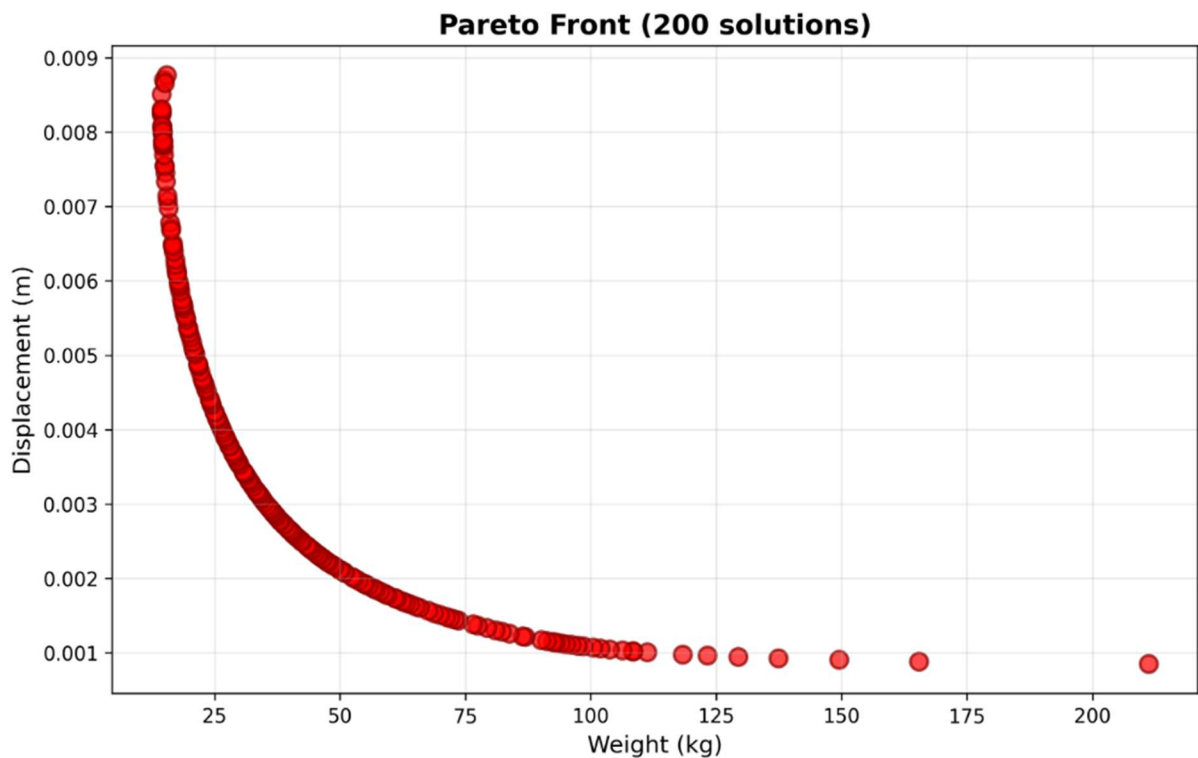
This linear distribution is used for exploring all the compromises between the two objectives:

- $w_1 \approx 1.0$ (strong weight) $\rightarrow$ Minimize the structural weight

- $w_1 \approx 0.5$ (equilibrze) → compromise solution
- $w_1 \approx 0.0$ (small weight) → Minimize the node displacement

Pareto Front

The Pareto front is the result of 200 runs of Simulated Annealing program and exhibits an optimal distribution of solutions representing the trade-off structural weight and node displacement.



The valid solutions are comprised between:

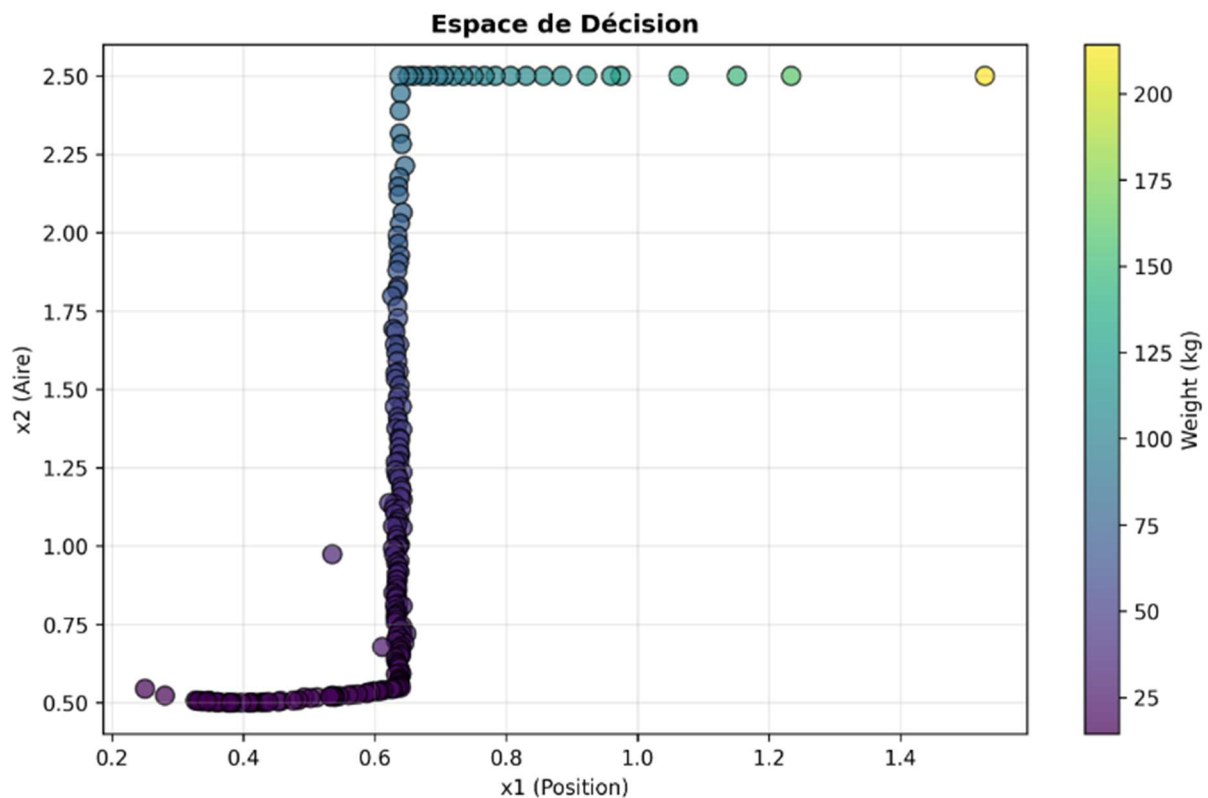
- **Weight:** 14.39 kg à 213.64 kg
- **Displacement :** 0.848 mm à 8.382 mm

The banana shape of the Pareto front saw the conflict between the structural optimization: reduce the weight and increase the flexibility and the displacement, otherwise if we want rigidity we need more material. We also saw no single solution minimizes in the same time both objectives.

Decision Space

The decision space distribution reveal a characteristic L shape. It is the result of the linear scalarization of the weight:

- **Region 1** ($x_2 \in [0,5; 1,0]$, x_1 variable): light solution for the weight minimization ($w_1 \approx 1,0$)
- **Region 2** ($x_2 \in [2,0; 2,5]$, x_1 variable): Minimize the node displacement ($w_1 \approx 0,0$)
- **Space of transition** ($x_2 \in [1,0 ; 2,0]$) : compromise solution



Analyse de Convergence

The convergence compartment of the 200 runs is represented on distinct plot on each objective:

Best weight evolution :

- Weight decreased to 200 kg by 14,39 kg.
- Significant improvement result after the 50 first runs
- Stable execution after 150 runs that saw a complete exploration of the light region

Best displament evolution :

- Speed convergence in 20 runs to the minimal value of 0,848 mm

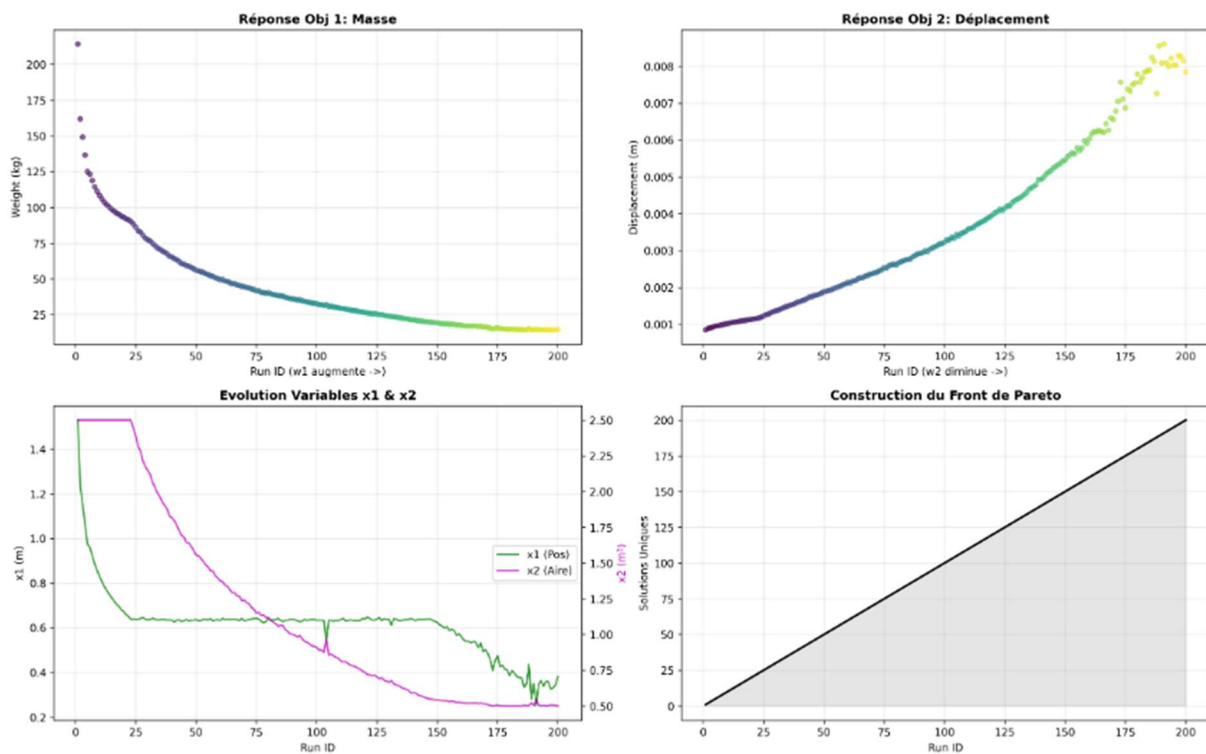
- We have a parabolic augmentation and this logic we start by making the displacement optimization, so we have the best optimization at the beginning and the worst after that.

Size of the Pareto front

- Linear augmentation: 1 solution added by execution (This is characteristic of the scalarization method). It is there for checking if we do not have a problem with the algorithm.
- So, a final size of 200 solutions (100% success)

Convergence Normalize:

- The objective of the weight is continually improving on the 200 runs
- The distance objective is fined on the first run
- The real best solution will be dependent on the context



TYPE	POIDS (kg)	DEPL. (mm)	x1 (Pos)	x2 (Aire)
1. MAX RIGIDITÉ	215.3952	0.8475	1.5350	2.5000
2. MAX LÉGÈRETÉ	14.3619	8.4318	0.3073	0.5112
3. COMPROMIS	32.3960	3.2627	0.6308	0.9027